

## 4.2.3 Plant tissues, organs and systems

### 4.2.3.1 Plant tissues

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Key opportunities for skills development                                    |
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| <p>Students should be able to explain how the structures of plant tissues are related to their functions.</p> <p>Plant tissues include:</p> <ul style="list-style-type: none"><li>• epidermal tissues</li><li>• palisade mesophyll</li><li>• spongy mesophyll</li><li>• xylem and phloem</li><li>• meristem tissue found at the growing tips of shoots and roots.</li></ul> <p>The leaf is a plant organ. Knowledge limited to epidermis, palisade and spongy mesophyll, xylem and phloem, and guard cells surrounding stomata.</p> | <p>AT 7</p> <p>Observation and drawing of a transverse section of leaf.</p> |

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| <p>Students should be able to explain how the structure of root hair cells, xylem and phloem are adapted to their functions.</p> <p>Students should be able to explain the effect of changing temperature, humidity, air movement and light intensity on the rate of transpiration.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>AT 3, 4, 5</p> <p>Measure the rate of transpiration by the uptake of water.</p> <p>AT 6, 7</p> <p>Investigate the distribution of stomata and guard cells.</p> <p>MS 2a, 2d, 5c</p> <p>Process data from investigations involving stomata and transpiration rates to find arithmetic means, understand the principles of sampling and calculate surface areas and volumes.</p> |
| <p>Students should be able to understand and use simple compound measures such as the rate of transpiration.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>MS 1a, 1c</p>                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Students should be able to:</p> <ul style="list-style-type: none"> <li>• translate information between graphical and numerical form</li> <li>• plot and draw appropriate graphs, selecting appropriate scales for axes</li> <li>• extract and interpret information from graphs, charts and tables.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>MS 2c, 4a, 4c</p>                                                                                                                                                                                                                                                                                                                                                              |
| <p>The roots, stem and leaves form a plant organ system for transport of substances around the plant.</p> <p>Students should be able to describe the process of transpiration and translocation, including the structure and function of the stomata.</p> <p>Root hair cells are adapted for the efficient uptake of water by osmosis, and mineral ions by active transport.</p> <p>Xylem tissue transports water and mineral ions from the roots to the stems and leaves. It is composed of hollow tubes strengthened by lignin adapted for the transport of water in the transpiration stream.</p> <p>The role of stomata and guard cells are to control gas exchange and water loss.</p> <p>Phloem tissue transports dissolved sugars from the leaves to the rest of the plant for immediate use or storage. The movement of food molecules through phloem tissue is called translocation.</p> <p>Phloem is composed of tubes of elongated cells. Cell sap can move from one phloem cell to the next through pores in the end walls.</p> <p>Detailed structure of phloem tissue or the mechanism of transport is not required.</p> |                                                                                                                                                                                                                                                                                                                                                                                   |